Assignment No 1

Aim: Identify the association between dependent and independent variables.

Theory:

In Artificial Intelligence and Machine Learning, a dataset consists of different variables or attributes that provide information about a particular problem. Before developing a machine learning model, it is important to understand how these variables are related to each other. The relationship between variables helps in identifying useful patterns and selecting appropriate features for model building.

A **dependent variable** is the variable whose value needs to be predicted, estimated, or explained by the machine learning model. It is also known as the **target variable, response variable, or output variable** and is usually represented by **Y**. For example, in a student performance dataset, the marks obtained by a student can be considered the dependent variable.

An **independent variable** is a variable that is used to explain or predict the dependent variable. It is also known as a **predictor, explanatory variable, input variable, or feature** and is represented by **X**. For example, study hours, attendance, and assignment scores can be independent variables used to predict a student's marks.

The **association between dependent and independent variables** refers to the degree to which a change in one variable is related to a change in another variable. Identifying this association helps us understand which independent variables have a meaningful relationship with the target variable.

Types of Association

The relationship between variables can generally be classified into the following types:

1. **Positive Association:**
   When the independent variable increases, the dependent variable also tends to increase.
   Example: More study hours may be associated with higher marks.

2. **Negative Association:**
   When the independent variable increases, the dependent variable tends to decrease.
   Example: As the number of errors increases, the overall quality of a product may decrease.

3. **No or Weak Association:**
   When changes in the independent variable do not show a clear relationship with the dependent variable.

Correlation

Correlation is one of the commonly used methods to measure the strength and direction of association between numerical variables. The **Pearson correlation coefficient** has a value between **-1 and +1**.

- **+1:** Perfect positive correlation
- **Values close to +1:** Strong positive association
- **0:** No linear correlation
- **Values close to -1:** Strong negative association
- **-1:** Perfect negative correlation

Correlation can be represented mathematically as:

$$r = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$$

where **Cov(X,Y)** is the covariance between X and Y, and **σX** and **σY** are the standard deviations of X and Y.

Methods Used to Identify Association

Different methods can be used depending on the type of variables:

- **Correlation analysis** – used for numerical variables.
- **Scatter plot** – visually represents the relationship between two numerical variables.
- **Heatmap** – displays correlation values among multiple numerical variables.
- **Covariance** – indicates the direction of relationship between variables.
- **Chi-square test** – used to determine association between categorical variables.
- **Statistical tests** – can be used to determine whether an observed relationship is statistically significant.

Importance in Machine Learning

Identifying the association between variables is an important part of **Exploratory Data Analysis (EDA)**. It helps to:

- Identify important features.
- Understand patterns in the dataset.
- Remove irrelevant or redundant variables.
- Detect highly correlated features.
- Improve feature selection.

- Understand the behavior of the target variable.
- Improve the performance and interpretability of machine learning models.

For example, suppose a dataset contains **Study Hours, Attendance, Assignment Score, and Final Marks**. Here, **Final Marks** can be the dependent variable, while **Study Hours, Attendance, and Assignment Score** can be independent variables. By calculating correlations and visualizing the relationships, we can determine which factors are more strongly associated with final marks.

Thus, identifying the association between dependent and independent variables is an important preliminary step in machine learning. It helps in understanding the dataset and selecting meaningful features before applying a machine learning algorithm.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

data = {
    "Hours_Studied": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    "Marks": [35, 40, 45, 52, 58, 65, 70, 78, 85, 92]
}

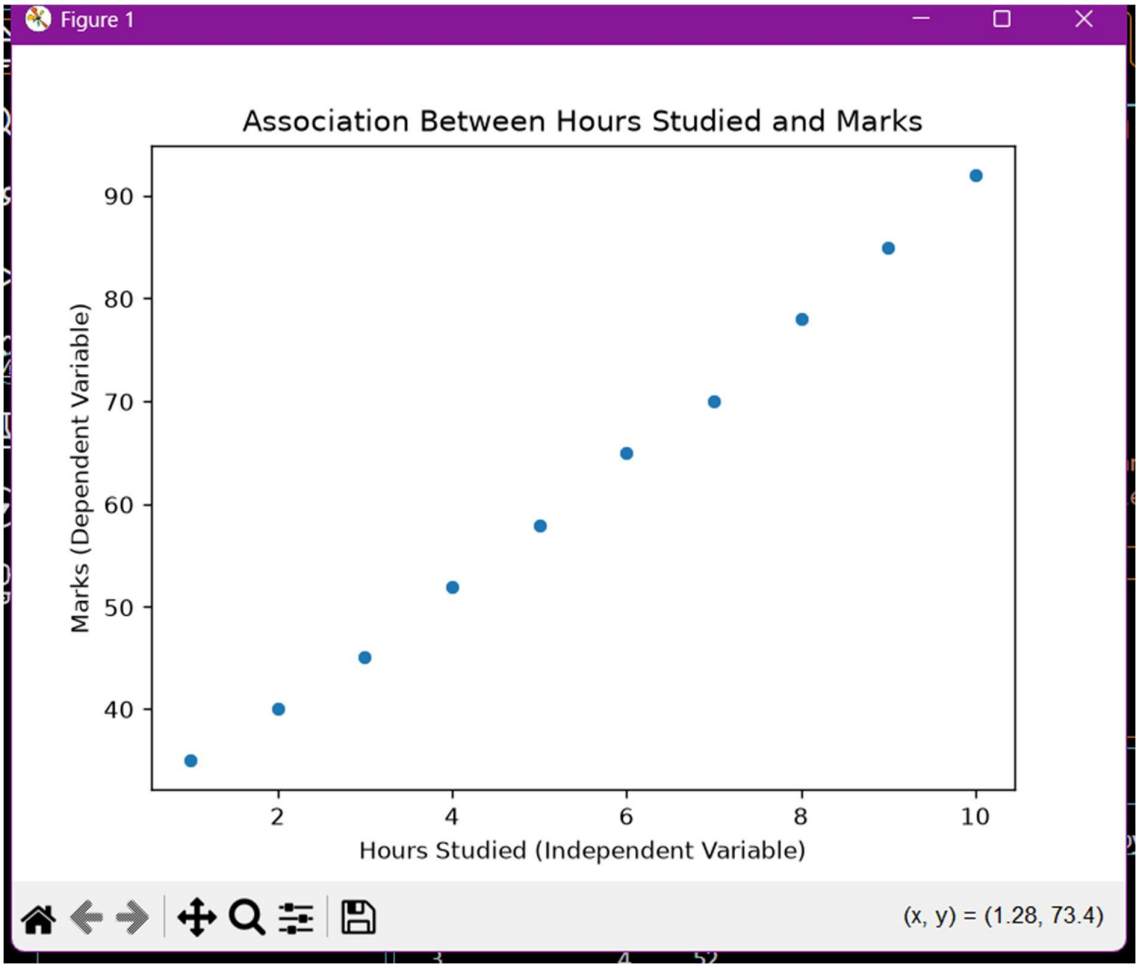
df = pd.DataFrame(data)

print("Dataset:")
print(df)

correlation = df["Hours_Studied"].corr(df["Marks"])
print("\nCorrelation between Hours Studied and Marks:")
print(correlation)

sns.scatterplot(
    x="Hours_Studied",
    y="Marks",
    data=df
)
plt.title("Association Between Hours Studied and Marks")
plt.xlabel("Hours Studied (Independent Variable)")
plt.ylabel("Marks (Dependent Variable)")
plt.show()
```

Output:




```
PS D:\CCTV_Object_Detection\pract1> python association.py
Dataset:
   Hours_Studied  Marks
0              1     35
1              2     40
2              3     45
3              4     52
4              5     58
5              6     65
6              7     70
7              8     78
8              9     85
9             10     92

Correlation between Hours Studied and Marks:
0.9985789846129846
```